Cornell University

CS 4782 – Spring 2026

Many Heads Are Better Than One

Reproducing MEDUSA – Speculative Decoding with Four Drafting Heads on Vicuna-7B

Huajie Zhong · Frank Dai

hz642 · sd924

1. Motivation

The problem. LLM inference is bottlenecked by *sequential* token generation: one token per forward pass. At batch size 1, GPUs are *memory-bandwidth-bound*, not compute-bound — most of the silicon is idle every step.

The goal. Generate multiple tokens per forward pass *without* sacrificing output quality.

MEDUSA [Cai et al., 2024] attaches K extra decoding heads to a frozen backbone, predicting tokens k steps ahead in parallel. A tree-attention *verification* pass checks all candidate continuations at once, and an acceptance rule guarantees quality.

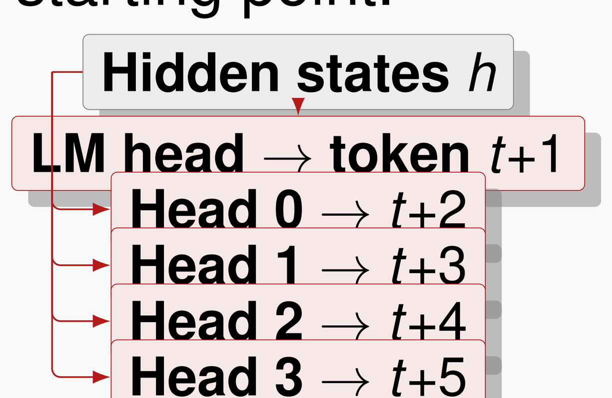
2. Architecture

MedusaHead: two-layer residual block

$$y = W_2\big(\mathrm{SiLU}(W_1 h) + h\big)$$

Key init trick: $W_1 = 0$, $W_2 =$ `lm_head.clone()`.

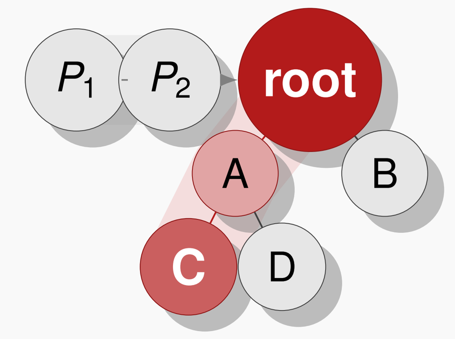
At step 0 each head *exactly* reproduces the original distribution — stable starting point.



Training (MEDUSA-1, frozen backbone):
K=4 heads only · loss $\mathcal{L} = \sum_{k=1}^{K} 0.8^k \mathcal{L}_k$ · 60k ShareGPT samples · AdamW 10^{-3} · H100 bf16.

3. Tree Attention

One forward pass verifies **64 candidate tokens**. Each node attends to prefix + ancestors only — not siblings, not cousins.



Mask for C: attends to prefix (P_1, P_2) + ancestors (root, A) + self; *not* siblings (B, D).

Static 64-node tree: widths $s = [10, 3, 2, 2]$, BFS-ordered.

Acceptance criteria:

Greedy: $\arg\max$ =child $\Rightarrow$ bitwise-identical to autoregressive baseline.

Typical (§2.3.1): $p(x) > \min(\varepsilon, \delta e^{-H(p)})$, $\varepsilon = \delta = 0.09$ – distributionally equivalent to sampling from the original.

4. Results

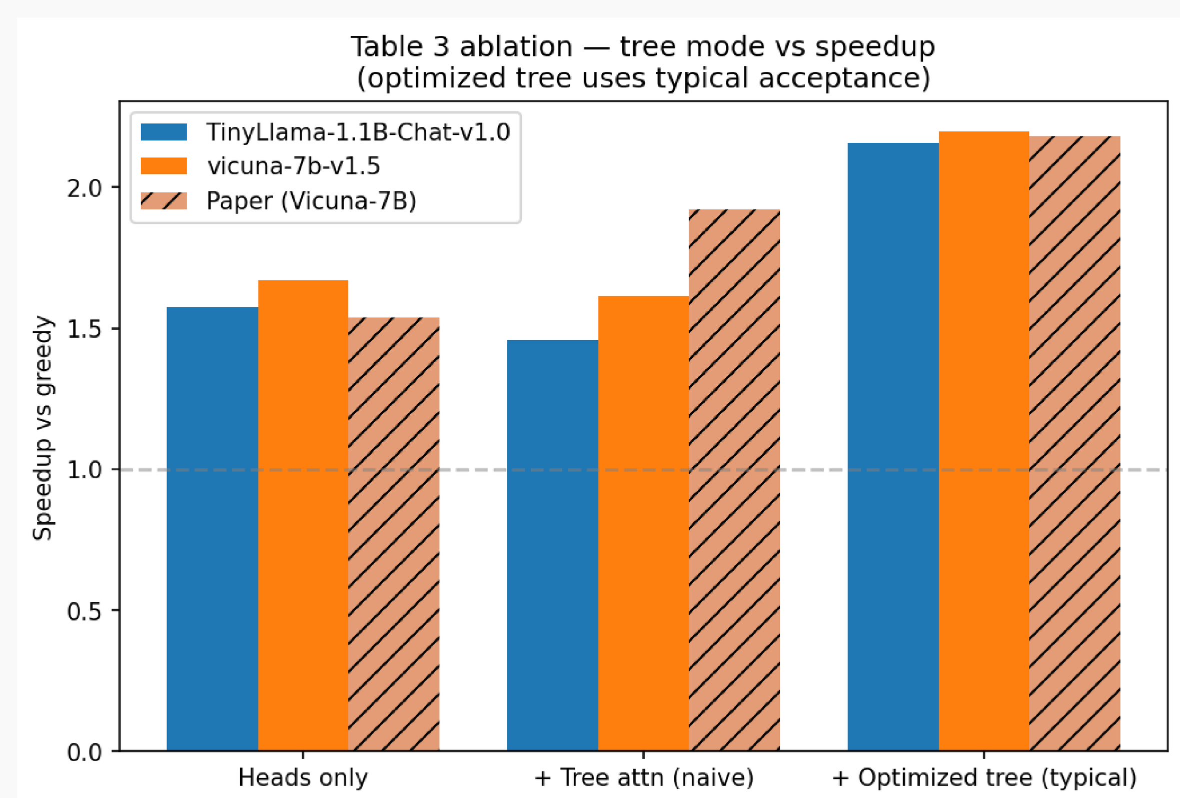


Figure 1: Table 3 ablation vs. paper targets.

Configuration	Paper	Ours
Heads only	1.54×	**1.67**×
+ Naive tree	1.92×	1.62×
+ Opt. tree (greedy)	—	2.034×
+ Opt. tree (typical)	**2.18**×	**2.196**× ✓
Head-0 accuracy	>60%	**64.1%**
Accept. length	3.47 (MED-2)	**2.87**

Table 1: Our reproduction matches the paper's 2.18× target.

5. Key Insight & Takeaways

Headline result.

2.196× speedup with typical acceptance on Vicuna-7B — matches the paper's **2.18**× target.

Surprise finding.

A *naive* tree (1.62×) is *slower* than no tree at all (1.67×): the dense 220-node Cartesian mask costs more than the +0.74 eff. tokens/step it buys (1.89 → 2.63). Tree-structure optimization is **load-bearing**, not cosmetic.

Typical > greedy acceptance.

+18% raw acceptance length (1.87 vs. 1.59 tree tokens/step), +11% effective (2.87 vs. 2.59), +7% TPS, with the §2.3.1 distributional-equivalence guarantee.

Gap analysis.

Our 2.87 eff. tokens/step sits below MEDUSA-2's 3.47 but meets the MEDUSA-1 <3.0 target. Residual gap: training on raw ShareGPT rather than Vicuna-regenerated responses.

Future work.

MEDUSA-2: joint fine-tuning with self-distillation (paper target ∼2.8×).

Regenerate training data from Vicuna-7B itself.

References
[1] Cai et al. MEDUSA. arXiv:2401.10774 (2024).
[2] Touvron et al. Llama 2. arXiv:2307.09288 (2023).
[3] Wolf et al. Transformers. EMNLP Demos (2020).
[4] Aeala. ShareGPT_Vicuna_unfiltered.

Code

Demo